



HORIZON EUROPE
Research & Innovation Framework



GENERATIVE BIONICS
Physical AI • Robotics Intelligence

GENERATIVE BIONICS

Analisi Strategica del Fit con i Bandi Horizon Europe
2026-2027

Data generazione: 15/04/2026 01:43

Cluster 4 – Digital, Industry and Space

Horizon Radar v1.1 • AHP + Gurobi LP

Executive Summary

FIT MEDIO CLUSTER

56.28%

CALL ANALIZZATE

15

Top 3 call

#1 • HORIZON-CL4-2026-02-DIGITAL-EMERGING-53-two-stage

[GO](#)

Innovative AI methods and technologies for the process industries (RIA) (Processes4Planet ...

63.8 / 100

#2 • HORIZON-CL4-2027-02-DIGITAL-EMERGING-52-two-stage

[GO](#)

New approaches for Human/AI collaboration for the workforce of the future (RIA) (Made in E...

62.8 / 100

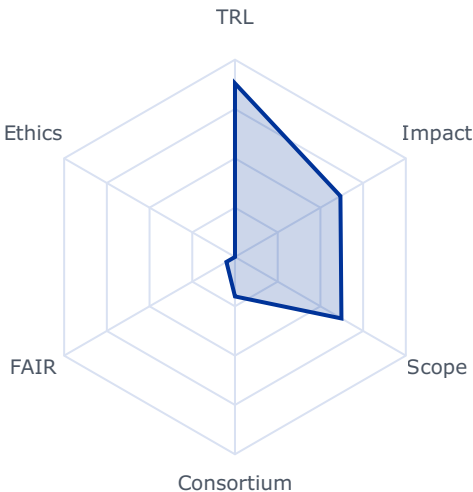
#3 • HORIZON-CL4-2026-02-DIGITAL-EMERGING-51-two-stage

[GO](#)

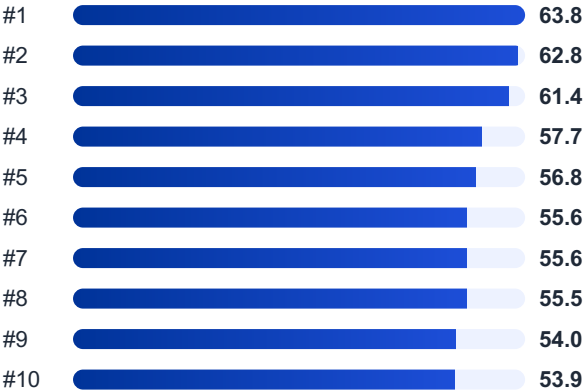
AI improved advanced manufacturing and production processes in factories (RIA) (Made in Eu...

61.4 / 100

Radar complessivo cluster



Distribuzione Top 10



Rank #1 • HORIZON-CL4-2026-02-DIGITAL-EMERGING-53-two-stage

GO

Innovative AI methods and technologies for the process industries (RIA) (Processes4Planet and AI, Data and Robotics partnerships)

Fit score: 63.84/100 • Solver: Optimal

Perché questa call ha preso 64/100

La call HORIZON-CL4-2026-02-DIGITAL-EMERGING-53-two-stage ottiene un fit del 63.8% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 32.2/100). Solver: Optimal, CR AHP: 0.0168.

*"Projects are expected to contribute to the following outcomes:"**"To develop and demonstrate innovative AI-driven solutions in materials and process"**"development to enhance the competitiveness and sustainability of the process industries"***Gap principali da colmare**

- ⚠ Nessun gap bloccante evidente dai vincoli hard del solver

Azioni concrete (prossimi 3-6 mesi)

- ✅ Consolidare la bozza di proposal con piano operativo e risk register

Spiegazione Tecnica AHP + LP

Matrice AHP (pairwise)

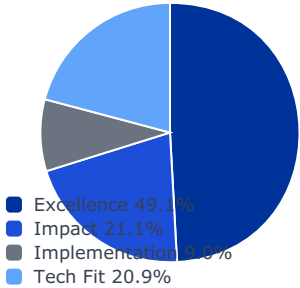
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0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

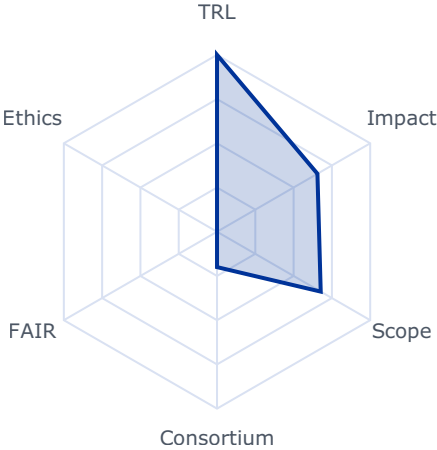
Criterio	Peso	Pre-LP	Post-LP	Breakdown
excellence	0.4914	0.6557	0.6557	0.3222
impact	0.2105	0.6613	0.6613	0.1392
implementation	0.0896	0.6800	0.4000	0.0358
tech_fit	0.2086	0.6770	0.6770	0.1412

CR: 0.0168 • Matrice consistente

Pesi finali AHP



Spider chart call



Vincoli LP applicati

- trl_violation: False
- budget_company_available: 1000000.0
- budget_max: 30000000.0
- sme_required: False
- sme_ok: True
- ssh_required: False
- gender_balance_required: False

Expected Outcomes

Projects are expected to contribute to the following outcomes:

- To develop and demonstrate innovative AI-driven solutions in materials and process development to enhance the competitiveness and sustainability of the process industries by better products and process technologies and reducing the time to market
- To increase the competitiveness of materials production in Europe by AI-supported optimal operation of plants and value networks and early detection of problems and failures
- To improve the working conditions in the plants by using AI technologies, metaverse, and robots.

137 This decision is available on the Funding and Tenders Portal, in the reference documents section for Horizon Europe, under 'Simplified costs decisions' or through this link:

https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/ls-decision_he_en.pdf

Scope

Drastically improved AI methods and technologies hold transformative potential for the process industries, enabling advancements in process design, operational efficiency, and faster innovation across the entire lifecycle of plants and products. By using different AI approaches such as multimodal generative AI, foundation models, and agentic AI, the industry can move beyond conventional AI applications as e.g. predictive maintenance and quality control toward more intelligent, adaptive, and creative solutions.

AI-based solutions can deliver value at every stage of the process lifecycle. In design and engineering, they can enable new innovations and accelerate the development process. During operations, AI technologies can be employed to optimize processes, enhance reproducibility, adapt to changing conditions, and provide new forms of support for the workforce and enable autonomous operation. In value chains, AI can help to adapt faster and detect changing customer needs. These capabilities support faster innovation and strengthen competitiveness in a rapidly evolving industrial landscape. However, realizing this potential requires careful consideration of risks related to reliability, security, and trust, ensuring that AI solutions are effective, safe, and responsible.

Proposals should produce dedicated innovative AI-based solutions for the process industry for one of the following scopes:

- More effective and faster development of new materials and processes
- Competitive and sustainable production, reducing the negative environmental impact of industry
- Reduction of risks for the health of the workforce and for the environment and making workplaces in the process industries more attractive.

In the projects, user acceptance and training of the users as well as integration into the OT/IT landscape of the companies should be taken into account, e.g. through active engagement in design, development and integration of systems and processes.

This topic is linked to the Apply AI Strategy, therefore proposals should seek collaboration with relevant initiatives.

Proposals should include a business case and exploitation strategy, as outlined in the introduction to Destination 'Leadership in materials and production for Europe'.

This topic implements the co-programmed European Partnerships Processes4Planet and AI, Data and Robotics.

Rank #2 • HORIZON-CL4-2027-02-DIGITAL-EMERGING-52-two-stage

GO

New approaches for Human/AI collaboration for the workforce of the future (RIA) (Made in Europe and AI, Data and Robotics partnerships)

Fit score: 62.82/100 • Solver: Optimal

Perché questa call ha preso 63/100

La call HORIZON-CL4-2027-02-DIGITAL-EMERGING-52-two-stage ottiene un fit del 62.8% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 33.8/100). Solver: Optimal, CR AHP: 0.0168.

"Industrial jobs are transformed through AI-based human-machine interactions (and skills"

"linked to them) which enhance flexibility, inclusion, well-being, up-skilling, career"

"Increased competitiveness and sustainability of advanced manufacturing industries by"

Gap principali da colmare

- ⚠ La call richiede copertura SSH esplicita

Azioni concrete (prossimi 3-6 mesi)

- ✅ Integrare partner SSH e work package dedicato a impatti sociali/adozione

Spiegazione Tecnica AHP + LP

Matrice AHP (pairwise)

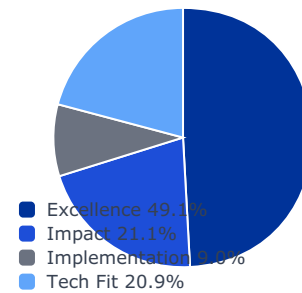
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

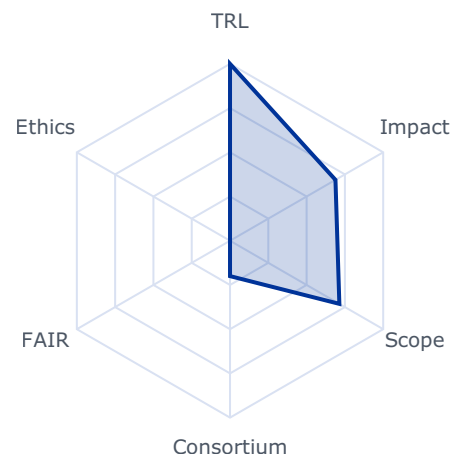
Criterio	Peso	Pre-LP	Post-LP	Breakdown
excellence	0.4914	0.6884	0.6884	0.3383
impact	0.2105	0.6722	0.5000	0.1052
implementation	0.0896	0.6800	0.4000	0.0358
tech_fit	0.2086	0.7139	0.7139	0.1489

CR: 0.0168 • Matrice consistente

Pesi finali AHP



Spider chart call



Vincoli LP applicati

- **trl_violation:** False
- **budget_company_available:** 1000000.0
- **budget_max:** 30000000.0
- **sme_required:** False
- **sme_ok:** True
- **ssh_required:** True
- **gender_balance_required:** False

Expected Outcomes

- Industrial jobs are transformed through AI-based human-machine interactions (and skills linked to them) which enhance flexibility, inclusion, well-being, up-skilling, career evolution and knowledge sharing;
- Increased competitiveness and sustainability of advanced manufacturing industries by means of knowledge formalization and adaptability of the machines to workers and markets based on different cultures.

Scope

Innovative AI approaches are poised to revolutionise human-machine collaboration in factories by fostering an environment where technology and human expertise synergistically enhance each other. AI can enhance the value of the companies by capturing and formalising the knowledge which is dispersed and not explicit. This allows companies to really own the knowledge and use it to reduce the onboarding time of new employees and support personnel upskilling to adapt to the evolving technological landscape. AI has a great potential to make task simpler by reducing the complexity offering intuitive interfaces and real-time feedback allowing workforce to be more efficient and effective while facilitating access to more complex tasks including those involving various forms of planning. AI can also adapt the interaction of automation with the worker taking into account particular needs of the human, including adaptation to the different abilities of the workers and facilitating inclusion. Finally, AI can be used to allow easier export of automation produced in EU by facilitating its interaction with workforce having different cultures adapting the interaction of the machines to the different needs.

Proposals should produce dedicated innovative AI approaches for human-machine collaboration in advanced manufacturing to be applied in at least two of following fields:

- Human-AI Co-Learning and knowledge capture to share competences, capture expert knowledge, provide interactive mentoring to up-skill the workforce, and support re-qualification and continuous training – leading to increased knowledge at factory level and avoiding loss of know-how.

- Human-AI teamwork thanks to innovative natural interaction models (considering the e.g. related hardware interfaces and/or collaborative machine tools), enabling to control

136 This decision is available on the Funding and Tenders Portal, in the reference documents section for Horizon Europe, under 'Simplified costs decisions' or through this link:

https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/ls-decision_he_en.pdf

complexity in cognitive cooperating production systems, including planning activities at shop floor level.

- Interfaces with automation which automatically adapt to the need of the humans including different abilities and different cultural needs.

Proposals should integrate a gender perspective and avoid any type of discrimination in the design and deployment of AI systems and human-machine interaction models, including addressing differences in user needs, such as needs of persons with disabilities, physical and cognitive ergonomics and training pathways. Proposals are also expected to identify and address other potential biases in AI systems to promote inclusive design that ensures safe and effective use by all workers. Human/AI collaboration requires utmost sensitivity to and consideration of human values and consideration of ethical principles as represented in Digital Humanism, therefore, appropriate consideration must also be given to the contribution of SSH.

This topic is linked to the Apply AI Strategy, therefore proposals should seek collaboration with relevant initiatives.

In addition, proposals are invited to build on the results of past projects on Extended Reality Technologies (XR), such as HORIZON-CL4-2021-HUMAN-01-13, HORIZON-CL4-2021-HUMAN-01-14, HORIZON-CL4-2021-HUMAN-01-25, HORIZON-CL4-2021-HUMAN-01-06, HORIZON-CL4-2021-HUMAN-01-28.

Proposals should include a business case and exploitation strategy, as outlined in the introduction to Destination 'Leadership in materials and production for Europe'.

This topic implements the co-programmed European Partnerships Made in Europe and AI, Data and Robotics.

Rank #3 • HORIZON-CL4-2026-02-DIGITAL-EMERGING-51-two-stage

GO

AI improved advanced manufacturing and production processes in factories (RIA) (Made in Europe and AI, Data and Robotics partnerships)

Fit score: 61.40/100 • Solver: Optimal

Perché questa call ha preso 61/100

La call HORIZON-CL4-2026-02-DIGITAL-EMERGING-51-two-stage ottiene un fit del 61.4% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 31.8/100). Solver: Optimal, CR AHP: 0.0168.

*"Increased competitiveness and productivity, through innovative AI-enabled advanced"**"manufacturing processes and operations, including real-time monitoring and adaptive"**"Reduction of emissions and alignment with Clean Industrial Deal objectives."***Gap principali da colmare**

- ⚠ Nessun gap bloccante evidente dai vincoli hard del solver

Azioni concrete (prossimi 3-6 mesi)

- ✅ Consolidare la bozza di proposal con piano operativo e risk register

Spiegazione Tecnica AHP + LP

Matrice AHP (pairwise)

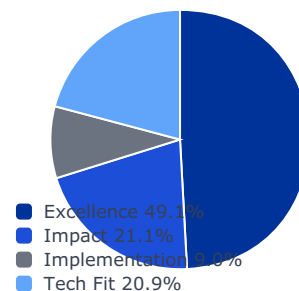
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0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

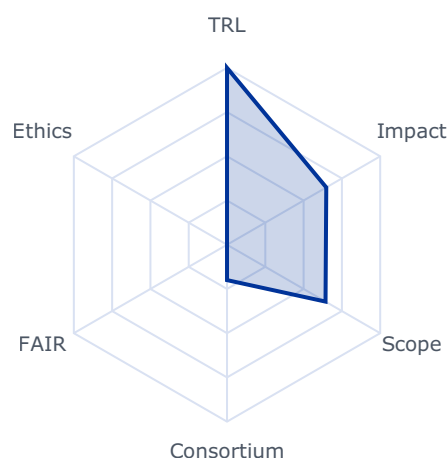
Criterio	Peso	Pre-LP	Post-LP	Breakdown
excellence	0.4914	0.6480	0.6480	0.3184
impact	0.2105	0.5969	0.5969	0.1256
implementation	0.0896	0.6800	0.4000	0.0358
tech_fit	0.2086	0.6430	0.6430	0.1341

CR: 0.0168 • Matrice consistente

Pesi finali AHP



Spider chart call



Vincoli LP applicati

- trl_violation: False
- budget_company_available: 1000000.0
- budget_max: 30000000.0
- sme_required: False
- sme_ok: True
- ssh_required: False
- gender_balance_required: False

Expected Outcomes

- Increased competitiveness and productivity, through innovative AI-enabled advanced manufacturing processes and operations, including real-time monitoring and adaptive optimisation; and
- Reduction of emissions and alignment with Clean Industrial Deal objectives.

Scope

AI approaches in manufacturing processes hold the potential to significantly enhance circularity, process and operational efficiency as well as sustainability of modern factories.

135 This decision is available on the Funding and Tenders Portal, in the reference documents section for Horizon Europe, under 'Simplified costs decisions' or through this link:

https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/lis-decision_he_en.pdf

Current state-of-the-art technologies have already paved the way for more streamlined operations, yet there remains untapped value in e.g. quality improvement, definition of optimal process operating conditions, reduction of scrap, optimization of energy usage. Real-time monitoring and adaptive optimisation using AI models can enable agile responses to production variability and support sustained, high-performance operations.

New solutions based on innovative enabling technologies such as deep learning, large language models, digital twins, synthetic data, and data-driven models allow manufacturers to improve production system efficiency, elevate product quality, and proactively address critical challenges in energy consumption and carbon footprint. This dual focus on operational excellence and sustainability ensures that factories can maintain competitive advantage while also contributing to specific environmental goals, e.g. reducing the pressure on ecosystems and natural resources. Since innovation capacity and competitiveness also requires a systemic understanding of an organization's value creating structure, novel AI solutions should be implemented such that they can support all structures and phases of operation, in technical and non-technical terms.

Proposals should produce dedicated innovative explainable AI based solutions in advanced manufacturing for at least two of the following:

- improve processes and operational efficiency, and reduce climate and environmental impact of processes and factories through dynamic selection of optimal processes and production parameters, exploiting AI for process modelling and/or optimisation;
- avoid the production of defective parts using AI to detect process drift and anomalies and correct proactively defects in real time; and
- maximise the fraction of regenerated components or materials used in the production using AI to optimise the material flow.

Proposals should demonstrate potential for these AI tools to adapt to changing production needs and real-time data, and should describe how industrial data access, confidentiality, and secure data-sharing will be addressed.

Projects are encouraged to link with AI Factories, including the Data Labs. The results may be validated in Testing and Experiment Facilities (TEFs), and further deployed via European Digital Innovation Hubs (EDIHs). This topic is linked to the Apply AI Strategy, therefore proposals should seek collaboration with relevant initiatives.

Proposals can optionally address the conditions for implementing the novel AI solutions within an organisations structure and value creating models, thereby contributing to systemic approach of implementing a smart organisation.

This topic is linked to the Apply AI Strategy, therefore proposals should seek collaboration with relevant initiatives.

Proposals should include a business case and exploitation strategy, as outlined in the introduction to Destination 'Leadership in materials and production for Europe'.

This topic implements the co-programmed European Partnerships Made in Europe and AI, Data and Robotics.

Rank #4 • HORIZON-CL4-2027-01-MAT-PROD-03

GO

Factory processes and automation for de- and re-manufacturing (RIA) (Made in Europe partnership)

Fit score: 57.70/100 • Solver: Optimal

Perché questa call ha preso 58/100

La call HORIZON-CL4-2027-01-MAT-PROD-03 ottiene un fit del 57.7% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 28.5/100). Solver: Optimal, CR AHP: 0.0168.

"A viable industrial ecosystem for circularity in manufacturing industries emerges,"

"De-manufacturing technologies and practices become available, making decisive"

"contributions to a European remanufacturing industry and market;"

Gap principali da colmare

- ⚠ Impact non ancora distintivo su outcomes e policy relevance

Azioni concrete (prossimi 3-6 mesi)

- ✅ Allineare proposition agli expected outcomes della call con KPI misurabili (60 giorni)

Spiegazione Tecnica AHP + LP

Matrice AHP (pairwise)

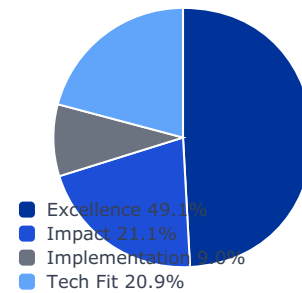
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0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

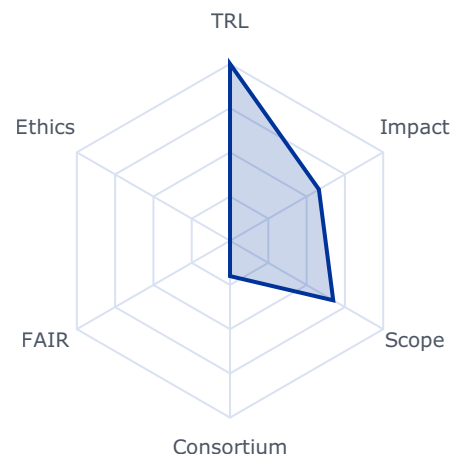
Criterio	Peso	Pre-LP	Post-LP	Breakdown
excellence	0.4914	0.5802	0.5802	0.2851
impact	0.2105	0.5491	0.5491	0.1156
implementation	0.0896	0.6800	0.4000	0.0358
tech_fit	0.2086	0.6738	0.6738	0.1406

CR: 0.0168 • Matrice consistente

Pesi finali AHP



Spider chart call



Vincoli LP applicati

- trl_violation: False
- budget_company_available: 1000000.0
- budget_max: 36000000.0
- sme_required: False
- sme_ok: True
- ssh_required: False
- gender_balance_required: False

Expected Outcomes

- A viable industrial ecosystem for circularity in manufacturing industries emerges, enhancing both circularity and resilience;
- De-manufacturing technologies and practices become available, making decisive contributions to a European remanufacturing industry and market;
- Functions of products are retained, reused, upgraded or adapted through de-manufacturing and re-manufacturing; and
- Skills, standards and safety measures relevant to remanufacturing are developed.

Scope

Proposals should focus on developing de-manufacturing and re-manufacturing technologies at the factory level, addressing at least three of the following:

- Technologies to efficiently analyse part condition and support predictive maintenance, including for re-manufactured parts or components of lower value, e.g. by combining multimodal sensor data, AI and human inputs;
- AI and robotic-assisted technologies, e.g. innovative end-effectors, to de-manufacture products and components, including handling, sorting and extended logistics;
- Model-based systems, to allow de-manufacturing and re-manufacturing operators to use CAD data and digital twins related to the original parts (and contribute to the development of a digital ecosystem);
- Solutions allowing local (on-site) repair or re-manufacturing of high-added value components (applied to e.g. wind turbines, aircraft and vessels); and
- Solutions to plan the sequence of operations based on the characteristics of the incoming products to be re-manufactured.

Proposals should include a business case and exploitation strategy, as outlined in the introduction to this Destination. Depending on the approaches to be pursued, these should consider the best solutions to enhance adoption, for example coupling manufacturing with de-/re- manufacturing plants, product and component value, risk mitigation for variable de-/re-manufacturing flows.

This topic implements the co-programmed European Partnership Made in Europe.

38 This decision is available on the Funding and Tenders Portal, in the reference documents section for Horizon Europe, under 'Simplified costs decisions' or through this link:

https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/ls-decision_he_en.pdf

Rank #5 • HORIZON-CL4-2026-01-MAT-PROD-23

GO

Accelerating the discovery and development of chemicals and innovative advanced materials through digitalisation and artificial intelligence (IA) (Innovative Advanced Materials for the EU partnership)

Fit score: 56.75/100 • Solver: Optimal

Perché questa call ha preso 57/100

La call HORIZON-CL4-2026-01-MAT-PROD-23 ottiene un fit del 56.8% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 29.1/100). Solver: Optimal, CR AHP: 0.0168.

"Accelerating the discovery and development process for innovative advanced materials"

"Innovation workflows which include design of experiment and/or design of simulation"

"Making a step change in the risk assessment of chemicals and advanced materials in"

Gap principali da colmare

- ⚠ Impact non ancora distintivo su outcomes e policy relevance

Azioni concrete (prossimi 3-6 mesi)

- ✅ Allineare proposition agli expected outcomes della call con KPI misurabili (60 giorni)

Spiegazione Tecnica AHP + LP

Matrice AHP (pairwise)

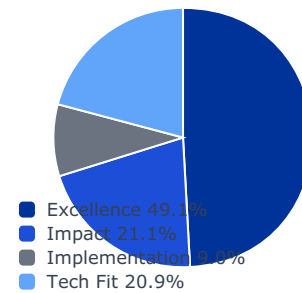
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

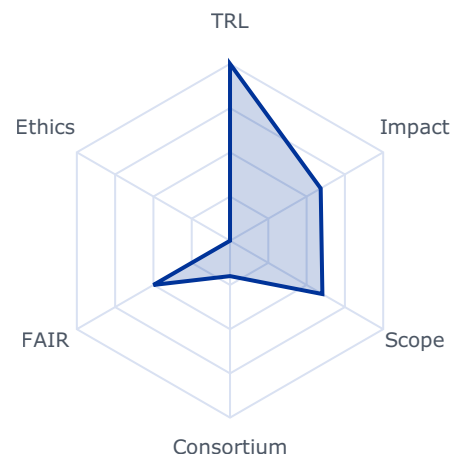
Criterio	Peso	Pre-LP	Post-LP	Breakdown
excellence	0.4914	0.5920	0.5920	0.2909
impact	0.2105	0.5462	0.5462	0.1150
implementation	0.0896	0.6800	0.4000	0.0358
tech_fit	0.2086	0.6030	0.6030	0.1258

CR: 0.0168 • Matrice consistente

Pesi finali AHP



Spider chart call



Vincoli LP applicati

- trl_violation: False
- budget_company_available: 1000000.0
- budget_max: 50000000.0
- sme_required: False
- sme_ok: True
- ssh_required: False
- gender_balance_required: False

Expected Outcomes

- Accelerating the discovery and development process for innovative advanced materials and chemicals through digital tools developed in Europe;
- Innovation workflows which include design of experiment and/or design of simulation
- Supporting the operationalisation of the SSbD framework;
- Making a step change in the risk assessment of chemicals and advanced materials in Europe.

Scope

Proposals should accelerate the pathway to market of new substances (chemicals or advanced materials) with superior or novel functionalities. This can be achieved with novel tools or proofs of concept using digital methods to accelerate development of new materials and demonstration of their properties. Where possible this should explore collaboration with other initiatives such as the Materials Commons for Europe or DIGIPASS, contributing data, modelling, digital tools applicable to the design, development, production, manufacturing, use and end of life phases, which connect to repeatable workflows. These workflows and tools may include the use of artificial intelligence as well as self-driving labs and their interconnection and design of experiment/design of simulation methods. They should also drive innovation in risk assessment, new test methods and support and facilitate the operationalisation and use of the SSbD framework³¹. Projects should include demonstrators which help to validate the materials development in realistic conditions.

By doing so, new innovative advanced materials (IAMs) with superior or novel functionalities and alternatives to substances of concern should be developed more rapidly in Europe. In addition, digital feedback loops ranging from requirements and information from production processes and scale-up, to manufacturing and integration into products, should be developed to accelerate market uptake. Innovative digital tools to speed up risk assessment and thereby market access of chemicals and advanced materials may also be addressed.

Interoperable workflows based on shared standards and dedicated ontologies, in particular through collaboration with the Materials Commons for Europe, should help to reduce the cost

³¹ See documents defining the SSbD framework on: https://ec.europa.eu/info/research-and-innovation/research-area/industrial-research-and-innovation/key-enabling-technologies/advanced-materials-and-chemicals_en

of the digital transition for industry with respect to circularity and safe and sustainable by design, e.g. by reducing the risk for adopters and vendors, and through modular tools that can be extended to new application domains without a major redesign. Tools should foster workflows in that ensure high-quality, well-structured and documented primary FAIR data and FAIR digital tools and workflows, enabling the re-use and/or streamlining of large data sets, facilitating academic and industrial collaborations and integrating AI and other digital technologies. Synergies with the SSbD toolboxes can also be foreseen including the adaptation and validation³² of the test methods for advanced materials. Proposals could also facilitate the generation of relevant data and where relevant sharing of data with the Common Data Platform for Chemicals. Where relevant, proposals should actively contribute to and cooperate with the EU Innovation and Substitution Hub(s). Proposals should allocate the necessary resources to the proposed activities.

Proposals should include a business case and exploitation strategy, as outlined in the introduction to this Destination.

Proposals could consider the involvement of the European Commission's Joint Research Centre (JRC), whose contribution could consist of added value to the operationalisation of the SSbD framework.

This topic implements the co-programmed European Partnership Innovative Advanced Materials for the EU (IAM4EU).

Rank #6 • HORIZON-CL4-2026-01-MAT-PROD-41

GO

Enhancing industry-academia knowledge exchange in Social Sciences and Humanities (SSH) (CSA)

Fit score: 55.62/100 • Solver: Optimal

Perché questa call ha preso 56/100

La call HORIZON-CL4-2026-01-MAT-PROD-41 ottiene un fit del 55.6% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 30.6/100). Solver: Optimal, CR AHP: 0.0168.

*"Increased innovation capabilities for industry by harnessing the potential of Social"**"Sciences and Humanities, including the Arts, to provide effective solutions to companies"**"research and innovation challenges and organisational development."***Gap principali da colmare**

- ⚠ Impact non ancora distintivo su outcomes e policy relevance
- ⚠ Implementation debole: piano esecutivo/consortium readiness non sufficiente
- ⚠ La call richiede copertura SSH esplicita

Azioni concrete (prossimi 3-6 mesi)

- ✅ Allineare proposition agli expected outcomes della call con KPI misurabili (60 giorni)
- ✅ Costruire shortlist partner (end-user, integrator, academia) e governance WP (90-120 giorni)
- ✅ Integrare partner SSH e work package dedicato a impatti sociali/adozione

Spiegazione Tecnica AHP + LP

Matrice AHP (pairwise)

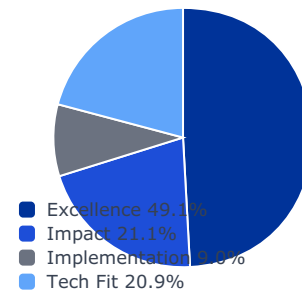
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

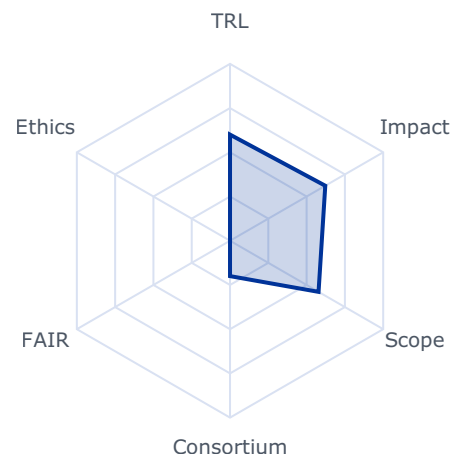
Criterio	Peso	Pre-LP	Post-LP	Breakdown
excellence	0.4914	0.6219	0.6219	0.3056
impact	0.2105	0.4476	0.4476	0.0942
implementation	0.0896	0.4400	0.4000	0.0358
tech_fit	0.2086	0.5780	0.5780	0.1206

CR: 0.0168 • Matrice consistente

Pesi finali AHP



Spider chart call



Vincoli LP applicati

- **trl_violation:** False
- **budget_company_available:** 1000000.0
- **budget_max:** 2000000.0
- **sme_required:** False
- **sme_ok:** True
- **ssh_required:** True
- **gender_balance_required:** False

Expected Outcomes

- Increased innovation capabilities for industry by harnessing the potential of Social Sciences and Humanities, including the Arts, to provide effective solutions to companies' research and innovation challenges and organisational development.
- Improved strategies to bring new products and technologies to the industry environment and ultimately to the market.
- By facilitating industry exposure, SSH researchers' better understanding of industry needs and opportunities for collaboration.

Scope

This action aims to leverage the strengths of social sciences, humanities and arts (SSH) to address companies' specific needs, fostering a dynamic and productive industry-academia co-creation for knowledge valorisation. This action will implement SSH–industry co-creation (for example hackathons, team-based approaches, targeted mentorship and exchange programmes etc) focussing on specific challenges from industry and SMEs including, but not limited to understanding the socio-technical implications of new technologies and innovations, broadening the perspectives of companies' strategic actions, creating a deeper understanding of customer needs, legal requirements and pathways to the market, strengthening the integration of social, economic and cultural inequality considerations into industry practices and innovation processes, developing new ideas and innovations and contributing to organisational development, sustainability and long-term profitability.

The action will cover the following activities:

50 This decision is available on the Funding and Tenders Portal, in the reference documents section for Horizon Europe, under 'Simplified costs decisions' or through this link:

https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/lis-decision_he_en.pdf

- Developing a methodology for understanding how various needs from industry, SMEs and customers can be addressed by knowledge exchanges with SSH researchers and students.
- Service to industry and SMEs including spinoffs and startups to support solving company challenges with international teams of SSH researchers and students.
- A study to tackle the key questions concerning the technical and conceptual feasibility of Industry-Academia knowledge exchange with SSH to improve innovation management and organisational development.

Rank #7 • HORIZON-CL4-2026-01-MAT-PROD-24

GO

Cooperation on innovative advanced materials with Japan (CSA)

Fit score: 55.56/100 • Solver: Optimal

Perché questa call ha preso 56/100

La call HORIZON-CL4-2026-01-MAT-PROD-24 ottiene un fit del 55.6% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 29.7/100). Solver: Optimal, CR AHP: 0.0168.

"Cooperation with Japan in the field of innovative advanced materials is strengthened."

"Advanced materials are an important factor for the competitiveness of Japanese and"

"European industries and are a crucial building block for strengthening resilience and open"

Gap principali da colmare

- ⚠ Impact non ancora distintivo su outcomes e policy relevance
- ⚠ Implementation debole: piano esecutivo/consortium readiness non sufficiente

Azioni concrete (prossimi 3-6 mesi)

- ✅ Allineare proposition agli expected outcomes della call con KPI misurabili (60 giorni)
- ✅ Costruire shortlist partner (end-user, integrator, academia) e governance WP (90-120 giorni)

Spiegazione Tecnica AHP + LP

Matrice AHP (pairwise)

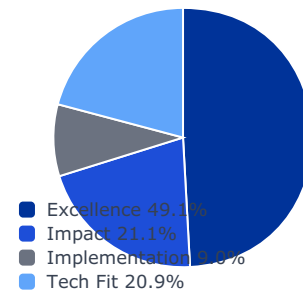
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

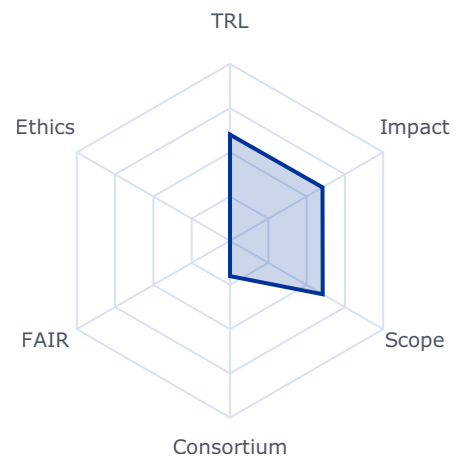
Criterio	Peso	Pre-LP	Post-LP	Breakdown
excellence	0.4914	0.6039	0.6039	0.2967
impact	0.2105	0.4430	0.4430	0.0932
implementation	0.0896	0.4400	0.4400	0.0394
tech_fit	0.2086	0.6051	0.6051	0.1262

CR: 0.0168 • Matrice consistente

Pesi finali AHP



Spider chart call



Vincoli LP applicati

- trl_violation: False
- budget_company_available: 1000000.0
- budget_max: 800000.0
- sme_required: False
- sme_ok: True
- ssh_required: False
- gender_balance_required: False

Expected Outcomes

- Cooperation with Japan in the field of innovative advanced materials is strengthened.

Scope

Advanced materials are an important factor for the competitiveness of Japanese and European industries and are a crucial building block for strengthening resilience and open strategic autonomy, including through international collaboration. Against this background, the European Commission and the Japanese Ministry of Science, Technology and Innovation Policy, have in April 2024 announced the launch of the EU-Japan Enhanced Dialogue³⁴ on Advanced Materials. This builds on the success of EU-Japan collaboration in R&I in material sciences. It aims to create a platform for sharing information on policy developments and exploring the opportunities to pursue collaborative research in the areas of mutual interest. As part of the EU-Japan Enhanced Dialogue on Advanced Materials and taking into account the Communication on Advanced Materials for Industrial Leadership³⁵, the purpose of this action is to enable European researchers in innovative advanced materials from Member States and Associated Countries to make research visits to related Japanese institutions. Proposals should aim to provide travel grants strengthening collaboration in relevant ongoing research activities in Europe on the topic of advanced materials with applications to either mobility, energy, construction, electronics or medical devices. Proposals should ensure a wide geographical coverage of the grants in Member States and Associated Countries. They should aim at visits to research groups at Japanese organisations with complementary activities for up to one month duration. Such travel grants should build on collaborations with Japanese organisations that are willing to receive such a visits. Proposals can either support the initiation of new collaborations, or existing collaborations.

Fast-tracking Circularity

Rank #8 • HORIZON-CL4-2027-01-MAT-PROD-61

GO

Fast Track to Research and Innovation for breakthroughs in industrial technologies (Research and Innovation Action)

Fit score: 55.55/100 • Solver: Optimal

Perché questa call ha preso 56/100

La call HORIZON-CL4-2027-01-MAT-PROD-61 ottiene un fit del 55.5% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 31.3/100). Solver: Optimal, CR AHP: 0.0168.

"Proposals should describe a credible pathway from their project results and outcomes"

"Proposals should enable open-ended breakthrough innovations in industrial technologies,"

"This is an open topic, intended to cover breakthrough innovations, up to TRL 4, within"

Gap principali da colmare

- ⚠ Impact non ancora distintivo su outcomes e policy relevance

Azioni concrete (prossimi 3-6 mesi)

- ✅ Allineare proposition agli expected outcomes della call con KPI misurabili (60 giorni)

Spiegazione Tecnica AHP + LP

Matrice AHP (pairwise)

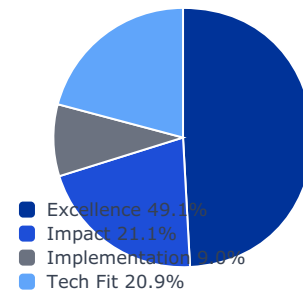
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

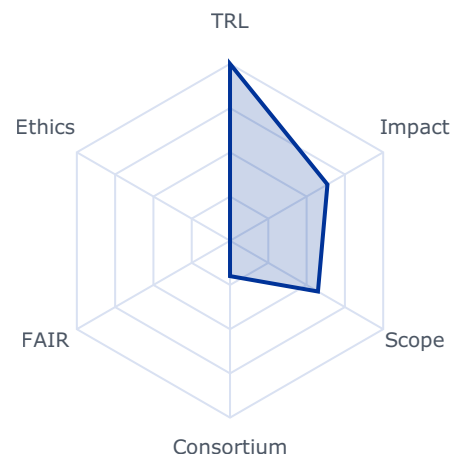
Criterio	Peso	Pre-LP	Post-LP	Breakdown
excellence	0.4914	0.6361	0.6361	0.3126
impact	0.2105	0.4154	0.4154	0.0874
implementation	0.0896	0.6800	0.4000	0.0358
tech_fit	0.2086	0.5740	0.5740	0.1197

CR: 0.0168 • Matrice consistente

Pesi finali AHP



Spider chart call



Vincoli LP applicati

- trl_violation: False
- budget_company_available: 1000000.0
- budget_max: 20000000.0
- sme_required: False
- sme_ok: True
- ssh_required: False
- gender_balance_required: False

Expected Outcomes

- Proposals should describe a credible pathway from their project results and outcomes towards the expected impact of this Destination;
- Proposals should enable open-ended breakthrough innovations in industrial technologies, feeding the pipeline of knowledge.

Scope

This is an open topic, intended to cover breakthrough innovations, up to TRL 4, within the scope of the Strategic Research and Innovation Agendas (SRIAs) of the partnerships Made in Europe, Process4Planet, Clean Steel, Innovative Advanced Materials for the EU

(IAM4EU) and Textiles for the Future.⁵² Proposals should carefully describe their expected contribution to one or more of the R&I objectives of the respective SRIA.

Rank #9 • HORIZON-CL4-2026-02-MAT-PROD-21-two-stage

WATCH

Development of safe and sustainable alternatives to substances of concern (IA)

Fit score: 54.05/100 • Solver: Optimal

Perché questa call ha preso 54/100

La call HORIZON-CL4-2026-02-MAT-PROD-21-two-stage ottiene un fit del 54.0% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 29.1/100). Solver: Optimal, CR AHP: 0.0168.

"Make safer and more sustainable alternatives to substances of concern available to the"

"industries offering products with targeted performances and supporting their"

"Speeding up the innovation cycle within a value chain important for European industry;"

Gap principali da colmare

- ⚠ Impact non ancora distintivo su outcomes e policy relevance

Azioni concrete (prossimi 3-6 mesi)

- ✅ Allineare proposition agli expected outcomes della call con KPI misurabili (60 giorni)

Spiegazione Tecnica AHP + LP

Matrice AHP (pairwise)

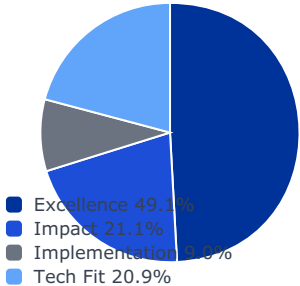
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

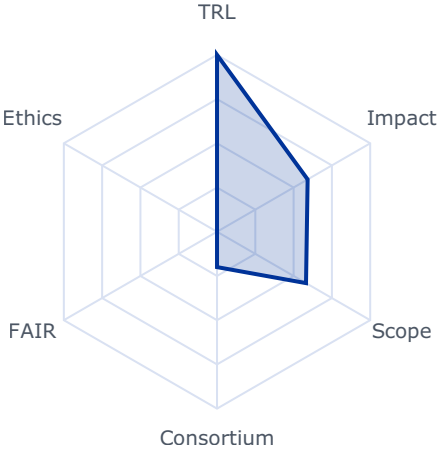
Criterio	Peso	Pre-LP	Post-LP	Breakdown
excellence	0.4914	0.5916	0.5916	0.2907
impact	0.2105	0.4408	0.4408	0.0928
implementation	0.0896	0.6800	0.4000	0.0358
tech_fit	0.2086	0.5810	0.5810	0.1212

CR: 0.0168 • Matrice consistente

Pesi finali AHP



Spider chart call



Vincoli LP applicati

- trl_violation: False
- budget_company_available: 1000000.0
- budget_max: 38000000.0
- sme_required: False
- sme_ok: True
- ssh_required: False
- gender_balance_required: False

Expected Outcomes

- Make safer and more sustainable alternatives to substances of concern available to the industries offering products with targeted performances and supporting their competitiveness,
- Speeding up the innovation cycle within a value chain important for European industry;
- Enhancing competitiveness of the industries by reducing regulatory and operational costs, while making supply chains more secure;
- Production processes, chemicals, materials and products that are inherently safer and more sustainable for a clean and autonomous economy; and
- Demonstrating how the safe and sustainable by design (SSbD) chemicals and materials framework can guide innovation and encourage innovators to minimise the use of substances of concern, reducing negative impacts on human health, climate and ecosystems.

Scope

The focus of this topic is on alternatives for the substitution of substances of concern (SoCs) as defined in the Ecodesign for Sustainable Product regulation²³. The design and development of these alternatives should lead to an innovation cycle covering their (re)design, development, production processes, and integration into products in manufacturing. The scope includes necessary developments of related processes and technologies to ensure alignment with and integration in industrial manufacturing facilitating uptake of the developed alternatives. If relevant, challenges for the adaption of existing production lines should be identified and solutions proposed.

Proposals should develop new chemical substances, innovative advanced materials or technologies to replace existing SoCs in one of the following areas: energy, mobility, construction, electronics, technical textiles as well as medical devices.

Proposals should demonstrate that the proposed alternative has a clear use case, market and potential to grow. The substitution barriers for the selected applications should be identified and effective mechanisms to maximise substitution within the targeted value chains proposed.

Proposals should demonstrate that SSbD framework ²⁴ will be applied throughout the innovation process, showing that safety and sustainability principles are actively integrated and influence decision-making in a transparent and traceable way, and ensure that the data generated within the proposal may be shared with the Common Data Platform for Chemicals. The new alternatives to be developed should meet the technical functions required in the specific applications while aligning their innovation process decision making with such framework.

Proposals are encouraged to cooperate with relevant projects and should contribute to and cooperate with the EU Innovation and Substitution Hub(s). Proposals should allocate the necessary resources to the proposed activities.

²³ REGULATION (EU) 2024/1781 establishing a framework for the setting of eco-design requirements for sustainable products, Article 2: definitions, recital 27

²⁴ See documents defining the SSbD framework and criteria on: https://ec.europa.eu/info/research-and-innovation/research-area/industrial-research-and-innovation/key-enabling-technologies/advanced-materials-and-chemicals_en

Proposals should include a business case and exploitation strategy, as outlined in the introduction to this Destination.

Proposals could consider involving the European Commission's Joint Research Centre (JRC), whose contribution could add value to the operationalisation of the SSbD framework.

Rank #10 • HORIZON-CL4-2026-01-MAT-PROD-46

WATCH

Mapping and service finder for Technology Infrastructures (CSA)

Fit score: 53.87/100 • Solver: Optimal

Perché questa call ha preso 54/100

La call HORIZON-CL4-2026-01-MAT-PROD-46 ottiene un fit del 53.9% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 27.2/100). Solver: Optimal, CR AHP: 0.0168.

"Improved visibility and access to technology infrastructures via an interactive portal"

"(open source, free to access, easy to update) of technology infrastructures across Europe;"

"55 This decision is available on the Funding and Tenders Portal, in the reference documents section for"

Gap principali da colmare

- ⚠ Impact non ancora distintivo su outcomes e policy relevance
- ⚠ Implementation debole: piano esecutivo/consortium readiness non sufficiente

Azioni concrete (prossimi 3-6 mesi)

- ✅ Allineare proposition agli expected outcomes della call con KPI misurabili (60 giorni)
- ✅ Costruire shortlist partner (end-user, integrator, academia) e governance WP (90-120 giorni)

Spiegazione Tecnica AHP + LP

Matrice AHP (pairwise)

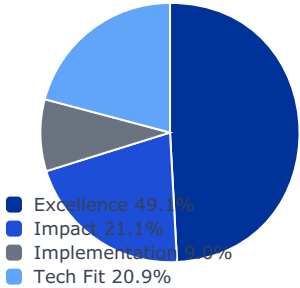
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0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

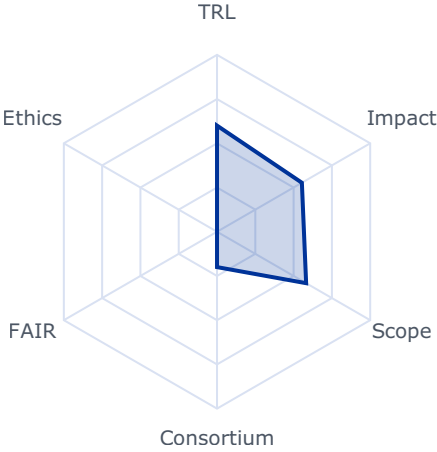
Criterio	Peso	Pre-LP	Post-LP	Breakdown
excellence	0.4914	0.5540	0.5540	0.2722
impact	0.2105	0.5193	0.5193	0.1093
implementation	0.0896	0.4400	0.4000	0.0358
tech_fit	0.2086	0.5817	0.5817	0.1213

CR: 0.0168 • Matrice consistente

Pesi finali AHP



Spider chart call



Vincoli LP applicati

- trl_violation: False
- budget_company_available: 1000000.0
- budget_max: 2000000.0
- sme_required: False
- sme_ok: True
- ssh_required: False
- gender_balance_required: False

Expected Outcomes

- Improved visibility and access to technology infrastructures via an interactive portal (open source, free to access, easy to update) of technology infrastructures across Europe;
55 This decision is available on the Funding and Tenders Portal, in the reference documents section for Horizon Europe, under 'Simplified costs decisions' or through this link:
https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/lis-decision_he_en.pdf
- Improved understanding of the European landscape of technology infrastructures, including existing facilities and their services supporting the development of strategic technologies by industrial users;
- Boosted innovation and market potential by providing an interactive portal where industrial users, especially SMEs, startups and scaleups, can search and find services relevant for their innovation and product development.

Scope

For many innovative startups, finding and accessing highly specific research and technology infrastructures services is a major hurdle. Information about the available services is limited, especially on infrastructures located beyond their local ecosystems.

Proposals are expected to create an inventory of existing and ongoing mapping of technology infrastructure and technology infrastructure services in Europe and to consolidate the knowledge into an interactive portal that will serve as technology infrastructure service finder. Furthermore, based on this inventory and other existing evidence-gathering and policy analysis, the proposals should use a comprehensive taxonomy of the technology infrastructures landscape. The portal, in line with regional smart specializations and existing ecosystems, should help assess which new technology infrastructure needs to be built to ensure long-term Europe's competitiveness and sovereignty.

Proposals should set out to establish an interactive portal including an interactive finder for technology infrastructure services in the EU. The finder should be aligned with other ongoing

Rank #11 • HORIZON-CL4-2026-01-MAT-PROD-11

WATCH

Innovative technologies and tools for exploration and data modelling of raw materials (RIA)

Fit score: 53.82/100 • Solver: Optimal

Perché questa call ha preso 54/100

La call HORIZON-CL4-2026-01-MAT-PROD-11 ottiene un fit del 53.8% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 27.7/100). Solver: Optimal, CR AHP: 0.0168.

"Increase information on the European Union raw materials, particularly critical raw"

"materials occurrences and deposits, while contributing to the implementation of the"

"National exploration programme,(article 19 of the Critical Raw Materials"

Gap principali da colmare

- ⚠ Impact non ancora distintivo su outcomes e policy relevance

Azioni concrete (prossimi 3-6 mesi)

- ✅ Allineare proposition agli expected outcomes della call con KPI misurabili (60 giorni)

Spiegazione Tecnica AHP + LP

Matrice AHP (pairwise)

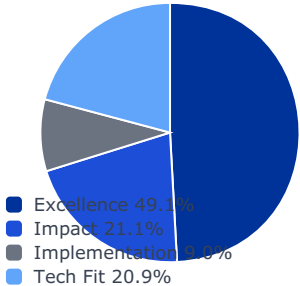
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

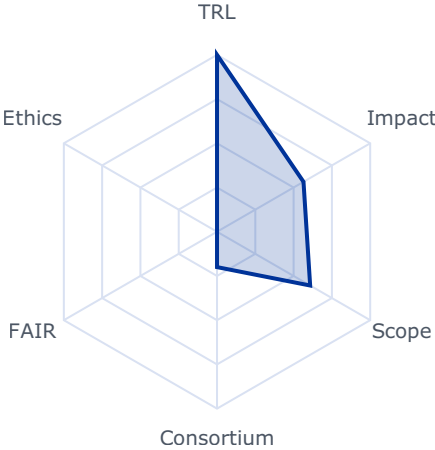
Criterio	Peso	Pre-LP	Post-LP	Breakdown
excellence	0.4914	0.5645	0.5645	0.2774
impact	0.2105	0.4665	0.4665	0.0982
implementation	0.0896	0.6800	0.4000	0.0358
tech_fit	0.2086	0.6081	0.6081	0.1268

CR: 0.0168 • Matrice consistente

Pesi finali AHP



Spider chart call



Vincoli LP applicati

- trl_violation: False
- budget_company_available: 1000000.0
- budget_max: 18500000.0
- sme_required: False
- sme_ok: True
- ssh_required: False
- gender_balance_required: False

Expected Outcomes

- Increase information on the European Union raw materials, particularly critical raw materials occurrences and deposits, while contributing to the implementation of the National exploration programme,(article 19 of the Critical Raw Materials Regulation(CRMA)63.
- Improve knowledge base of EU raw materials to identify new areas for exploration and resource estimation;
- Accelerate development of EU domestic raw materials exploration projects integrating innovative technologies that can form the basis for new EU SMEs;
- Develop innovative exploration data acquisition, processing and modelling and mineral system analysis for identification of critical raw materials deposits in the EU;
- Projects will provide technologies and data which will strengthen EU Geological Surveys capacities and skills to implement the National Exploration programmes as defined in the CRMA.
- Accelerate development of EU domestic raw materials exploration projects by junior mining / exploration companies.

Scope

Actions should develop and validate advanced geological modelling and mineral system analysis using multi-source data (geological, geophysical, and geochemical) from ground-based and remote-sensing techniques to develop high-resolution 3D models of mineral deposits. The integration of new (AI and machine learning) and conventional methods will be necessary to predict with the greatest accuracy the location of mineral deposits of critical raw materials and their carrier minerals. Actions contributing to the National exploration programmes under the article 19 of CRMA is encouraged.

Actions should develop new knowledge and conceptual models, supported by innovative technologies to strengthen and secure the EU's supply of primary raw materials by:

- Generating better geological understanding (i.e. characterization, modelling, mapping) of known mineral deposits to facilitate discovery of new resources, including mineral systems carrying critical raw materials;

62 This decision is available on the Funding and Tenders Portal, in the reference documents section for Horizon Europe, under 'Simplified costs decisions' or through this link:

https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/ls-decision_he_en.pdf

63 Regulation (EU) 2024/1252 of the European Parliament and of the Council of 11 April 2024 establishing a framework for ensuring a secure and sustainable supply of critical raw materials and amending Regulations (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1724 and (EU) 2019/1020

- Collecting new geological, geophysical, and geochemical data and developing ore genetic models and mineral system analysis to build a broad understanding of EU's deposit type, including critical raw materials in order to identify areas for exploration, especially in previously underexplored regions;
- Advancing mineral prospectivity modelling processes;
- Facilitate the integration of existing multi-datasets with newly acquired data.

Proposals should include a business case and exploitation strategy, as outlined in the introduction to this Destination.

Rank #12 • HORIZON-CL4-2027-01-MAT-PROD-02

WATCH

Advanced manufacturing for key products (IA) (Made in Europe partnership)

Fit score: 53.70/100 • Solver: Optimal

Perché questa call ha preso 54/100

La call HORIZON-CL4-2027-01-MAT-PROD-02 ottiene un fit del 53.7% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 27.6/100). Solver: Optimal, CR AHP: 0.0168.

"Advanced manufacturing technology and machinery becomes available in Europe for the"

"36 The production of batteries (including for automotive use) is addressed by topic HORIZON-CL5-2026"

"05-D2-03: Integrated Production and Product Development for Next-Generation Lithium-based"

Gap principali da colmare

- ⚠ Impact non ancora distintivo su outcomes e policy relevance

Azioni concrete (prossimi 3-6 mesi)

- ✅ Allineare proposition agli expected outcomes della call con KPI misurabili (60 giorni)

Spiegazione Tecnica AHP + LP

Matrice AHP (pairwise)

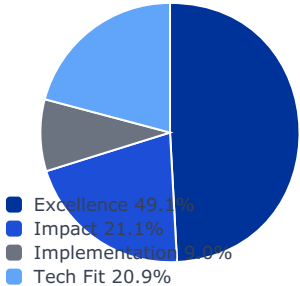
1.000	3.000	5.000	2.000
0.333	1.000	3.000	1.000
0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

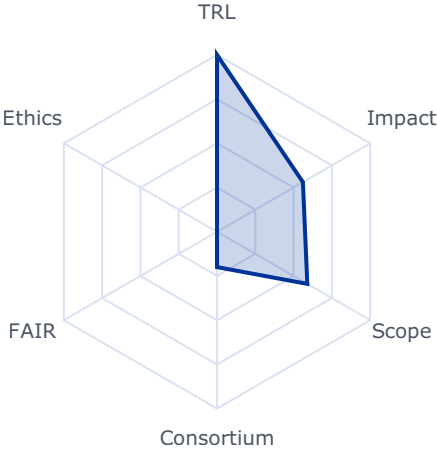
Criterio	Peso	Pre-LP	Post-LP	Breakdown
excellence	0.4914	0.5607	0.5607	0.2755
impact	0.2105	0.4881	0.4881	0.1027
implementation	0.0896	0.6800	0.4000	0.0358
tech_fit	0.2086	0.5895	0.5895	0.1230

CR: 0.0168 • Matrice consistente

Pesi finali AHP



Spider chart call



Vincoli LP applicati

- trl_violation: False
- budget_company_available: 1000000.0
- budget_max: 36000000.0
- sme_required: False
- sme_ok: True
- ssh_required: False
- gender_balance_required: False

Expected Outcomes

- Advanced manufacturing technology and machinery becomes available in Europe for the manufacturing of key and high-performance products;
- 36 The production of batteries (including for automotive use) is addressed by topic HORIZON-CL5-2026-05-D2-03: Integrated Production and Product Development for Next-Generation Lithium-based Batteries for Mobility.
- Where relevant, production becomes increasingly circular through the reuse of secondary raw materials; and/or innovative advanced materials are incorporated in manufactured products, leading to better performance and quality;
- Resource efficiency in terms of materials and energy is increased significantly; and
- Circularity, productivity and competitiveness are increased, and hence resilience of European industry is enhanced.

Scope

This topic addresses technologies and machinery for advanced manufacturing, focusing on manufacturing excellence and on increasing circularity, including through the better use of innovative advanced materials and secondary raw materials. The focus is on key manufactured components and products that are competitive and have enhanced performance, and contribute to Europe's technological leadership in manufacturing, but which are at risk of being lost to Europe or rely on raw materials or parts whose supply is mostly coming from outside Europe.

Proposals should develop technologies and machinery to enable the manufacturing of these components with a minimal use of critical raw materials [reference to overall targets] or imported materials. This includes an increased use of secondary raw materials or biobased materials or revalorised components.

Where appropriate to enhance performance and quality, proposals should target the use of innovative advanced materials (such as lightweight, functionalised or self-healing materials). In this case, the development of the advanced materials should not be the main focus of proposals, nevertheless the necessary steps to adapt such innovative advanced materials to the needs of the manufacturing application should be included. These can include digital twins for materials as well as SSbD-design steps.

Examples of advanced manufacturing technologies and machinery include, but are not restricted to:

- Innovative additive manufacturing;
- Hybrid manufacturing (additive, subtractive);
- Photonics;
- Advanced joining technologies;
- Polymer composite manufacturing;
- Advanced technologies for surface treatment and structuring, to tailor surface properties for specific applications;
- Manufacturing of components with lightweight materials; and
- In-line testing.

Proposals should include a business case and exploitation strategy, as outlined in the introduction to this Destination.

Applications in the automotive industry may be considered. However, the production of batteries is not within the scope of this topic.³⁷

International cooperation is encouraged, especially with Japan or Taiwan.

This topic implements the co-programmed European Partnership Made in Europe.

Rank #13 • HORIZON-CL4-2027-01-MAT-PROD-22

WATCH

Innovative advanced materials and new production processes – reducing dependencies on Critical and Strategic Raw Materials (IA) (Innovative Advanced Materials for the EU and Processes4Planet partnerships)

Fit score: 53.48/100 • Solver: Optimal

Perché questa call ha preso 53/100

La call HORIZON-CL4-2027-01-MAT-PROD-22 ottiene un fit del 53.5% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 28.6/100). Solver: Optimal, CR AHP: 0.0168.

"Reducing dependencies of critical and strategic raw materials through partial or total"

"substitution by safe and sustainable innovative advanced materials and/or via more"

"efficient use of critical and strategic raw materials in production processes;"

Gap principali da colmare

- ⚠ Impact non ancora distintivo su outcomes e policy relevance

Azioni concrete (prossimi 3-6 mesi)

- ✅ Allineare proposition agli expected outcomes della call con KPI misurabili (60 giorni)

Spiegazione Tecnica AHP + LP

Matrice AHP (pairwise)

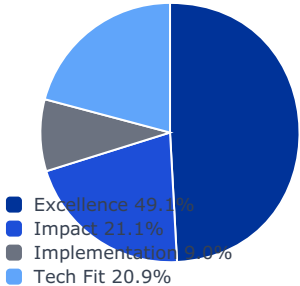
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0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

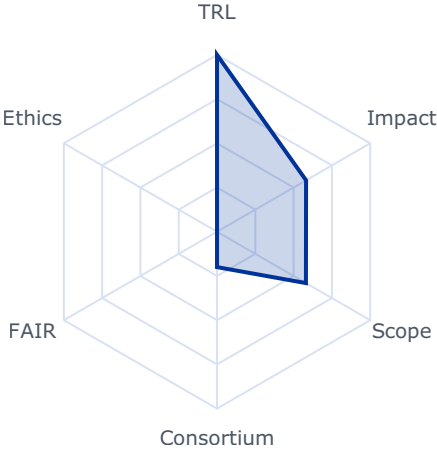
Criterio	Peso	Pre-LP	Post-LP	Breakdown
excellence	0.4914	0.5814	0.5814	0.2857
impact	0.2105	0.4383	0.4383	0.0923
implementation	0.0896	0.6800	0.4000	0.0358
tech_fit	0.2086	0.5805	0.5805	0.1211

CR: 0.0168 • Matrice consistente

Pesi finali AHP



Spider chart call



Vincoli LP applicati

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- budget_max: 36000000.0
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- ssh_required: False
- gender_balance_required: False

Expected Outcomes

- Reducing dependencies of critical and strategic raw materials through partial or total substitution by safe and sustainable innovative advanced materials and/or via more efficient use of critical and strategic raw materials in production processes;
 - Speeding up the innovation cycle within value chain(s) important for European industry;
 - Enhancing competitiveness of the industries and operational costs, while making supply chains more secure;
 - New or improved production processes, innovative advanced materials and products that are safer and more sustainable, supporting a clean and autonomous economy; and
- 26 https://single-market-economy.ec.europa.eu/sectors/raw-materials/areas-specific-interest/raw-materials-diplomacy_en
- 27 https://policy.trade.ec.europa.eu/eu-trade-relationships-country-and-region/negotiations-and-agreements_en
- Demonstrating how the safe and sustainable by design (SSbD) chemicals and materials framework can guide innovation.

Scope

The focus of this topic is on alternatives for the substitution or more efficient use of critical and strategic raw materials²⁸. The design and development of innovative advanced materials (IAMs) and processes should lead to an innovation cycle covering the (re)design of materials and production processes, and the integration of IAMs into products. Proposals should develop IAMs or process technologies to replace or reduce the use of critical and strategic raw materials in strategic areas and sectors such as energy, mobility, construction, electronics, medical devices or chemical industries.

Proposals should address one or several of the following approaches:

- Design, development and production with targets on performance, safety and sustainability of IAMs substituting or making a more efficient use of critical and strategic raw materials.
- Innovative industrial processes for the reduction of the use of critical and strategic raw materials focussed on optimizing process safety, sustainability, flexibility, scalability, cost-efficiency.
- Co-development strategies for IAMs and industrial processes. These strategies should demonstrate the value of co-development through specific use cases while maintaining broad relevance across various materials and process types.

Proposals should demonstrate clear use case(s), market and potential to grow. The substitution barriers for the selected applications should be identified and a driving mechanism for a maximal substitution in the targeted value chains proposed.

The scope includes necessary adaptations of related processes and technologies to ensure alignment with and integration in industrial manufacturing in order to facilitate the uptake of the developed solutions. If relevant, challenges for the adaption of existing production lines should be identified and solutions proposed.

Proposals should demonstrate that SSbD framework 29 will be applied throughout the innovation process, showing that safety and sustainability principles are actively integrated and influence decision-making in a transparent and traceable way, and ensure that the data generated within the proposal may be shared with the Common Data Platform for Chemicals.

The new alternatives to be developed should meet the technical functions required in the specific applications while aligning their innovation process decision making with such framework.

²⁸ <https://rmis.jrc.ec.europa.eu/eu-critical-raw-materials>; as well as Annex I and II of the

²⁹ See documents defining the SSbD framework on: https://ec.europa.eu/info/research-and-innovation/research-area/industrial-research-and-innovation/key-enabling-technologies/advanced-materials-and-chemicals_en

Proposals should include a business case and exploitation strategy, as outlined in the introduction to this Destination.

Proposals are encouraged to cooperate with relevant projects. Where relevant, proposals should actively contribute to and cooperate with the EU Innovation and Substitution Hub(s)³⁰.

Proposals should allocate the necessary resources to the proposed activities

This topic implements the co-programmed European partnerships Innovative Advanced Materials for the EU (IAM4EU) and Processes4Planet.

Rank #14 • HORIZON-CL4-2026-01-MAT-PROD-44

WATCH

Attracting management talent for capacity building for Technology Infrastructures staff members (CSA)

Fit score: 53.19/100 • Solver: Optimal

Perché questa call ha preso 53/100

La call HORIZON-CL4-2026-01-MAT-PROD-44 ottiene un fit del 53.2% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 28.8/100). Solver: Optimal, CR AHP: 0.0168.

"Proposals should demonstrate the following expected outcomes:"

"Sound understanding of the competences and skills needs of Technology Infrastructures"

"related to effective management and providing services to industry in all member states;"

Gap principali da colmare

- ⚠ Impact non ancora distintivo su outcomes e policy relevance
- ⚠ Implementation debole: piano esecutivo/consortium readiness non sufficiente

Azioni concrete (prossimi 3-6 mesi)

- ✅ Allineare proposition agli expected outcomes della call con KPI misurabili (60 giorni)
- ✅ Costruire shortlist partner (end-user, integrator, academia) e governance WP (90-120 giorni)

Spiegazione Tecnica AHP + LP

Matrice AHP (pairwise)

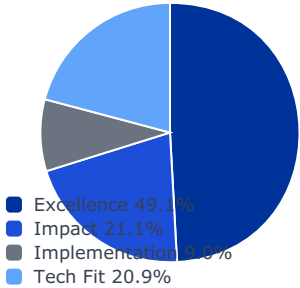
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0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

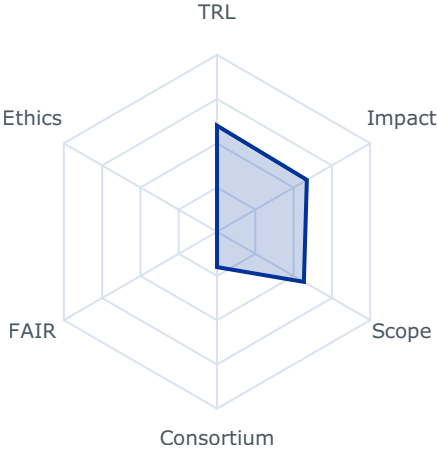
Criterio	Peso	Pre-LP	Post-LP	Breakdown
excellence	0.4914	0.5865	0.5865	0.2882
impact	0.2105	0.4257	0.4257	0.0896
implementation	0.0896	0.4400	0.4000	0.0358
tech_fit	0.2086	0.5670	0.5670	0.1183

CR: 0.0168 • Matrice consistente

Pesi finali AHP



Spider chart call



Vincoli LP applicati

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Expected Outcomes

Proposals should demonstrate the following expected outcomes:

- Sound understanding of the competences and skills needs of Technology Infrastructures related to effective management and providing services to industry in all member states;
- Improved and structure the training of Technology Infrastructures staff members with improved sector-agnostic competencies;
- Increased and improved collaboration between Research and Technology Infrastructures, leading to exchange of best practices in management and capacity building in Technology Infrastructures, and peer learning;
- Better uptake of technology infrastructure services by industry, including startups and scaleups, innovators and researchers;
- Improved coherence, visibility and accessibility of technology infrastructures facilities and services;
- Improved business and governance models of Technology Infrastructures.

Scope

A combination of factors including rapid technological advancements, inefficient strategic planning, skills gaps, and budget limitations can create sub-optimal technology infrastructure management. Proposals are expected to develop and provide schemes to attract management talent (including but not limited to organisational sciences) to technology infrastructures in EU member states. These management talent schemes should be aimed at attracting talents that will study the technology infrastructures needs, in terms of competencies, skills, organizations, and business models, to deliver more efficiently their services to industry, especially to SMEs, startups and scaleups. Based on this sound understanding of Technology Infrastructures needs, the attracted management talents are expected to both develop and provide advice to Technology Infrastructures senior management and trainings to their staff members aimed at increasing their capacity to develop and provide services to industry. The developed trainings should be sector-agnostic, open-source, accessible remotely, easy to update, and take stock of other existing similar initiatives and experiences in Research and Technology Infrastructures.

Proposals should propose ways to enhance skills and career profiles of technical staff working in technology infrastructures to address evolving needs such as research security, data management, quality assurance, etc., as well as further professionalising the training of managerial and leadership staff in technology infrastructures. They should help also to promote entrepreneurial skills for technology infrastructures staff in training curricula to fully capture the potential of technology infrastructures as centres of deep-tech innovation ecosystem.

https://ec.europa.eu/info/funding-tenders/opportunities/docs/2021-2027/horizon/guidance/ls-decision_he_en.pdf

Furthermore, proposals should aim to enhance transnational and multisite collaboration among Technology Infrastructures fostering the building of technology infrastructures ecosystems, the sharing of experience, resources, and best practices in management and capacity building, staff exchange and networking. These collaborations could lay the foundations of a future framework to facilitate coordination of technology infrastructures policy and priority setting at EU level as well as exchange of experience and good practices. Proposals may also benefit, especially for communication and dissemination activities, from sharing best practices and taking stock of what other relevant projects are developing in the field of Technology Infrastructures. Existing specific best management practices and European frameworks in the technology fields considered by the proposals have to be taken into account.

Rank #15 • HORIZON-CL4-2026-01-MAT-PROD-01

WATCH

Advanced manufacturing for key products (IA) (Made in Europe partnership)

Fit score: 52.85/100 • Solver: Optimal

Perché questa call ha preso 53/100

La call HORIZON-CL4-2026-01-MAT-PROD-01 ottiene un fit del 52.9% dopo ottimizzazione LP. Il criterio dominante e' excellence (contributo 26.8/100). Solver: Optimal, CR AHP: 0.0168.

"Advanced manufacturing technology and machinery becomes available in Europe for the"

"Where relevant, production becomes increasingly circular through the reuse of"

"secondary raw materials; and/or innovative advanced materials are incorporated in"

Gap principali da colmare

- ⚠ Excellence sotto soglia competitiva: narrativa scientifica da rafforzare
- ⚠ Impact non ancora distintivo su outcomes e policy relevance

Azioni concrete (prossimi 3-6 mesi)

- ✅ Definire proof plan tecnico con milestone TRL e metriche di validazione (90 giorni)
- ✅ Allineare proposition agli expected outcomes della call con KPI misurabili (60 giorni)

Spiegazione Tecnica AHP + LP

Matrice AHP (pairwise)

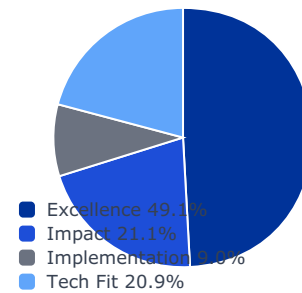
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0.200	0.333	1.000	0.500
0.500	1.000	2.000	1.000

Pre-LP vs Post-LP

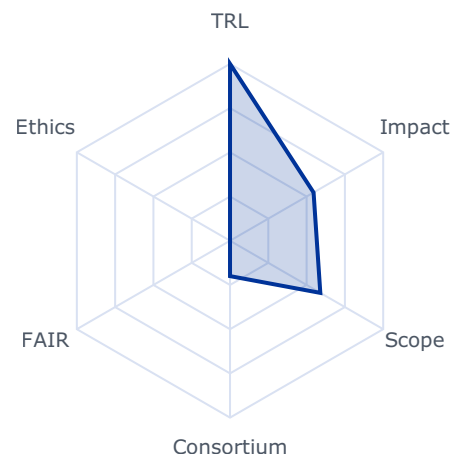
Criterio	Peso	Pre-LP	Post-LP	Breakdown
excellence	0.4914	0.5455	0.5455	0.2680
impact	0.2105	0.4830	0.4830	0.1017
implementation	0.0896	0.6800	0.4000	0.0358
tech_fit	0.2086	0.5895	0.5895	0.1230

CR: 0.0168 • Matrice consistente

Pesi finali AHP



Spider chart call



Vincoli LP applicati

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- **ssh_required:** False
- **gender_balance_required:** False

Expected Outcomes

- Advanced manufacturing technology and machinery becomes available in Europe for the manufacturing of key and high-performance products;
- Where relevant, production becomes increasingly circular through the reuse of secondary raw materials; and/or innovative advanced materials are incorporated in manufactured products, leading to better performance and quality;
- Resource efficiency in terms of materials and energy is increased significantly; and
- Circularity, productivity and competitiveness are increased, and hence resilience of European industry is enhanced.

Scope

This topic addresses technologies and machinery for advanced manufacturing, focusing on manufacturing excellence and on increasing circularity, including through the better use of innovative advanced materials and secondary raw materials. The focus is on key manufactured components and products that are competitive and have enhanced performance, and contribute to Europe's technological leadership in manufacturing, but which are at risk of being lost to Europe or rely on raw materials or parts whose supply is mostly coming from outside Europe.

Proposals should develop technologies and machinery to enable the manufacturing of these components with a minimal use of critical raw materials [reference to overall targets] or imported materials. This includes an increased use of secondary raw materials or biobased materials or revalorised components.

Where appropriate to enhance performance and quality, proposals should target the use of innovative advanced materials (such as lightweight, functionalised or self-healing materials).

In this case, the development of the advanced materials should not be the main focus of proposals, nevertheless the necessary steps to adapt such innovative advanced materials to the needs of the manufacturing application should be included. These can include digital twins for materials as well as SSbD-design steps.

Examples of advanced manufacturing technologies and machinery include, but are not restricted to:

- Innovative additive manufacturing;
- Hybrid manufacturing (additive, subtractive);
- Photonics;
- Advanced joining technologies;
- Polymer composite manufacturing;
- Advanced technologies for surface treatment and structuring, to tailor surface properties for specific applications;
- Manufacturing of components with lightweight materials; and
- In-line testing.

Proposals should include a business case and exploitation strategy, as outlined in the introduction to this Destination.

The portfolio approach will be used, to ensure that at least one proposal focusing on the automotive industry is funded. However, the production of batteries is not within the scope of this topic.³⁶

International cooperation is encouraged, especially with Japan or Taiwan.

This topic implements the co-programmed European Partnership Made in Europe.

Disclaimer

Report generato automaticamente da Horizon Radar v1.0 – Strumento decisionale basato su Analytic Hierarchy Process (AHP) + Linear Programming (Gurobi). I risultati sono indicativi e supportano la prioritizzazione strategica.

Generative Bionics R&D Team



13/04/2026 01:43
Horizon Radar